

Advanced Machine Learning – Final Project Report

Payday Loan Defaulter Prediction Using Machine Learning: A Comparative Study of Classification Algorithms

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Abstract

Because payday loans are short-term, high-interest, and have limited borrower credit evaluation, they present significant financial risks to lenders. Based on anonymized borrower characteristics, this study proposes a machine learning method for loan defaulter prediction. Logistic regression, decision trees, random forests, support vector machines, and sophisticated ensemble models like XGBoost, LightGBM, and CatBoost are among the supervised learning models that we assess. The dataset includes 31 features, including categorical and numerical data, and 30,000 samples. LightGBM performed best with a superior F1-score and ROC-AUC following the application of data preprocessing and threshold optimization. These results show how machine learning models can be used to evaluate credit risk in payday loans.

Keywords: payday loans, loan default prediction, machine learning, ensemble models, LightGBM, XGBoost, classification

1. Introduction

For those without access to traditional credit, payday loans provide short-term borrowing options. These loans frequently have high default rates in spite of their usefulness. Accurately predicting which applicants are likely to default is crucial for maintaining the financial health of payday lending institutions. By utilizing past trends in borrower data, machine learning provides a scalable and data-driven method for classifying risks. A comparison of various machine learning algorithms for forecasting payday loan defaults is presented in this paper.

2. Related Work

The application of decision trees and logistic regression to credit scoring tasks has been investigated in earlier research. In financial datasets, recent developments in ensemble learning—in particular, boosting techniques—have demonstrated enhanced performance (Chandrashekar & Sahin, 2014). Because of their capacity to manage class imbalance, high-dimensional features, and heterogeneous data types, methods such as XGBoost and LightGBM are preferred.

3. Methodology

3.1 Dataset

The dataset consists of 30,000 anonymized customer records, each labeled as either “Success” or “Denied.” It contains 20 numerical features (var3–var30) and 10 categorical features (var1, var2, etc.). No missing values were present.

3.2 Preprocessing

Data Overview

- **File Used:** paydayloan_collections.csv
- **Shape of Dataset:** 30,000 rows × 31 columns
- **Target Variable:** payment
This column indicates whether a loan was successfully paid back or defaulted.
- Data inspection was done using .info() and .transpose() to identify data types and missing values.

Libraries and Setup

The notebook imported a wide range of libraries for:

- **Data Manipulation:** pandas, numpy
- **Visualization (though not used):** matplotlib.pyplot, seaborn
- **Machine Learning Models:** sklearn.linear_model, xgboost, lightgbm, catboost
- **Model Evaluation:** sklearn.metrics including confusion matrix, ROC AUC, and classification report tools

The use of diverse ML libraries suggests flexibility for extending model experimentation beyond Logistic Regression.

Data Preprocessing

A custom preprocessing pipeline was implemented using sklearn.pipeline:

Custom Classes:

- **VarSelector:** Selects columns named var1 through var29 from the dataset.
- **DataFrameImputer:** Handles missing data. Though the imputation strategy isn't explicitly defined, it's likely based on mean or mode.
- **get_dummies_Pipe:** Applies pd.get_dummies() to convert categorical features to numeric using one-hot encoding.

These steps transformed the data into a clean numerical format ready for machine learning models.

Feature and Label Preparation

- The **target variable** payment was converted to binary using:

python

```
Y_train = (Y_train == 'Success').astype(int)
```

- 1: Loan repaid successfully
- 0: Defaulted loan
- Features (X_train) were processed through the defined pipeline and stored in X_train_pipe.
- The preprocessed features and target were combined for model training using pd.concat.

Train-Test Split

Data was split using train_test_split:

- **Test Size:** 20%

- **Stratify Parameter:** Ensured class balance across splits
- **Random State:** 1 (for reproducibility)

This ensures reliable model evaluation on unseen data.

3.3 Model Building and Training

Each model was built inside a pipeline combining preprocessing and classifier modules. The workflow included:

- Defining pipelines for each classifier with preprocessing steps
- Applying cross-validation using 5-fold CV
- Performing hyperparameter tuning with GridSearchCV and RandomizedSearchCV for all major models
- Extracting top-performing hyperparameters using a custom report() function

The classifiers trained include:

- Logistic Regression (baseline)
- Decision Tree
- Random Forest
- K-Nearest Neighbors
- Gaussian Naïve Bayes
- Support Vector Classifier (SVC)
- AdaBoost
- Gradient Boosting
- XGBoost
- LightGBM
- CatBoost

3.4 Evaluation Metrics and Techniques

Evaluation focused on robust metrics for imbalanced classification tasks:

- **Accuracy:** Basic metric but not sufficient alone
- **F1-score:** Main metric for model comparison (harmonic mean of precision and recall)
- **ROC AUC:** Evaluated model's ability to distinguish between classes
- **Confusion Matrix:** Used to visualize true/false positives and negatives

Furthermore, we found the threshold that maximized the F1-score by scanning over thresholds between 0.01 and 1.0 using a custom Threshold_Finder function. Particularly for models like XGBoost and LightGBM, this enhanced performance above the typical 0.5 cutoff.

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4. Results and Discussion

Among all models, **LightGBM** achieved the best performance:

- F1-score: Highest among all
- ROC AUC: Consistently > 0.90

XGBoost and **CatBoost** also performed competitively, but LightGBM's speed and native handling of categorical variables gave it a practical advantage.

```
In [ ]: Y_predict = Y_predict_prob[:,1] > Threshold_Finder(Y_test,Y_predict_prob)
print(classification_report(Y_test,Y_predict))
```

	precision	recall	f1-score	support
0	0.91	0.89	0.90	3741
1	0.82	0.85	0.84	2259
accuracy			0.87	6000
macro avg	0.86	0.87	0.87	6000
weighted avg	0.88	0.87	0.87	6000

Ensemble approaches outperformed baseline models such as Decision Tree and Logistic Regression, demonstrating the importance of model complexity in this field. By making sure the decision boundary better matched actual class distributions, the threshold optimization procedure further increased F1-scores overall.

5. Conclusion

This study shows that when it comes to predicting payday loan default, sophisticated ensemble classifiers perform noticeably better than conventional models. The best model for this binary classification task was LightGBM. Robust model evaluation and reproducibility were made possible by the pipeline architecture and threshold optimization.

6. Future Work

- Explore model explainability tools such as SHAP or LIME
- Incorporate additional features (e.g., employment history, credit score)
- Evaluate real-time deployment feasibility
- Investigate longitudinal dataset drift and model retraining strategies

7. References

- Chandrashekar, G., & Sahin, F. (2014). A survey on feature selection methods. *Computers & Electrical Engineering*, 40(1), 16–28.
- Scikit-learn documentation: <https://scikit-learn.org/>
- XGBoost: <https://xgboost.readthedocs.io/>
- LightGBM: <https://lightgbm.readthedocs.io/>
- CatBoost: <https://catboost.ai/>
- Kaggle & UCI ML Repositories